

OSCAR BARNES

omcbarnes@gmail.com | oscarbarnes.github.io | linkedin.com/in/omcb1

TECHNICAL SKILLS

Proficient Languages: Python, R, SQL, Excel

Tools & Libraries: Sklearn, Pandas, NumPy, ggplot, MySQL, Git, LaTeX

Machine Learning & Stats: KNN, K-Means Clustering, PCA, GLMs, EM-Algorithm, MCMC Sampling

EDUCATION

Master of Mathematics and Statistics (MMathStat) | University of Warwick

Jul 2027

Grade: 1st Class Honours (74.6% overall so far)

Honours & Awards: Statistics Senior Scholarship (Awarded for 3rd-Year Academic Merit; 75.6% Average)

Master's Dissertation: Applications of Extreme Value Theory (EVT) for Interval-Censored Athlete Data

Relevant Modules: Maths of Machine Learning, Machine Learning Frameworks, Multivariate Statistics, Generalised Linear Models, Python for Data-Analytic Tasks, Applied Statistical Modelling, Probability Theory

A-levels | Notre Dame High School – Sheffield

Sep 2023

Grades: Mathematics A*, Further Mathematics A*, Economics A

Honours & Awards: Departmental Mathematics Prize (Showing Dedication, Commitment and Contribution)

PROJECTS

2026 FIFA World Cup Monte Carlo Simulator | Python / Streamlit

- Built a Hierarchical Bayesian Negative Binomial model in PyMC to estimate latent attacking and defending strengths across 284 teams, applying MCMC sampling and Laplace-smoothed penalty priors.
- Simulated 1,000 World Cup brackets (104,000+ matches) to compute knockout survival probabilities, deploying an interactive Streamlit dashboard to analyse match overdispersion and extreme tail events.

Premier League Stratification Analysis | R / Excel / LaTeX

- Engineered a machine learning pipeline to challenge the “Big 6” narrative by extracting and analysing multi-dimensional team performance metrics.
- Applied Principal Component Analysis (PCA) for dimensionality reduction and K-Means clustering to identify hidden similarities and group teams into distinct statistical profiles.

Car Price Prediction Engine | R / LaTeX

- Conducted Exploratory Data Analysis (EDA) on a real-world Kaggle dataset, utilising K-Nearest Neighbours (KNN) to robustly impute missing data points.
- Developed and tuned Ridge, Lasso and Stepwise regression models to predict car prices, evaluating accuracy and preventing overfitting using R^2 and Adjusted R^2 metrics.

EXPERIENCE

Barnett Waddingham (now Howden)

Leeds, UK

Actuarial Consulting Intern

July – Aug 2025

- Supported actuarial valuations for pension schemes with 5,000+ members by validating 10+ cash equivalent transfer values and reconciling member data using Excel.
- Produced and updated multiple client actuarial reports and accounting disclosures, ensuring accuracy and delivering work to tight deadlines. Drafted professional client communications and presented funding valuations to mock trustees.

Sheffield Scitt

Sheffield, UK

Maths Teacher Intern

Jun – Jul 2024

- Applied diverse teaching techniques to maintain engagement across varied ability groups.
- Led tutor groups of 5-10 students, supporting a 50% improvement in student test scores.

INTERESTS

- Active member of the University Statistics Football team (automated an Excel fantasy league data pipeline), keen squash player.
- Grade 5 pianist focusing on learning cinematic film scores, enthusiastic home cook who enjoys trying new recipes.